

Exploring Spotify Song Popularity Through Audio Features

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Discussion Article

For my final project, I wanted to explore what makes songs popular. Using a Spotify dataset and a Shiny app, I was able to create something that allows users to see relationships between different song characteristics and popularity. Users can select a genre, choose a song feature such as danceability, energy, loudness, acousticness, and more, and filter songs by maximum length. The app then displays a scatterplot showing the relationship between the selected feature and popularity and then it calculates the correlation between the two variables, and displays the most popular songs that match the selected filters.

One of the most interesting findings from this project was that the relationship between popularity and musical characteristics often depended on the genre being examined. When looking at genres such as Hip-Hop and Pop, features like danceability and energy generally showed positive relationships with popularity. This suggests that songs that are easier to dance to or have higher energy levels may be more likely to become popular within these genres. However, the strength of these relationships was often weaker than expected. While many people may assume that a single feature strongly determines popularity, the correlations were usually small to moderate, showing that popularity is influenced by many factors beyond the audio features included in the dataset.

Another interesting finding was the role of song length. By allowing users to filter songs based on duration, it became possible to compare shorter and longer songs within the same genre. In many cases, shorter songs appeared more frequently among the most popular tracks. This may reflect modern listening habits, where shorter songs are more likely to be replayed or streamed multiple times. However, this pattern was not consistent across every genre. For example, some classical pieces and other instrumental works remained popular despite being much longer than the average song.

The project also highlighted how different genres have distinct musical characteristics. For example, Classical music generally showed lower danceability values than genres such as Hip-Hop or Pop, while acousticness tended to be much higher. These differences help explain why

a feature that is associated with popularity in one genre may not have the same relationship in another. Comparing genres demonstrated that there is no universal formula for creating a popular song.

Overall, this project showed that song popularity is a complex outcome that cannot be explained by a single audio category. Instead, popularity appears to result from a combination of musical characteristics, genre-specific trends, and likely many outside influences that are not captured in the dataset. The Shiny app made it possible to explore these relationships interactively and gain a better understanding of how different musical features contribute to the success of songs on Spotify.

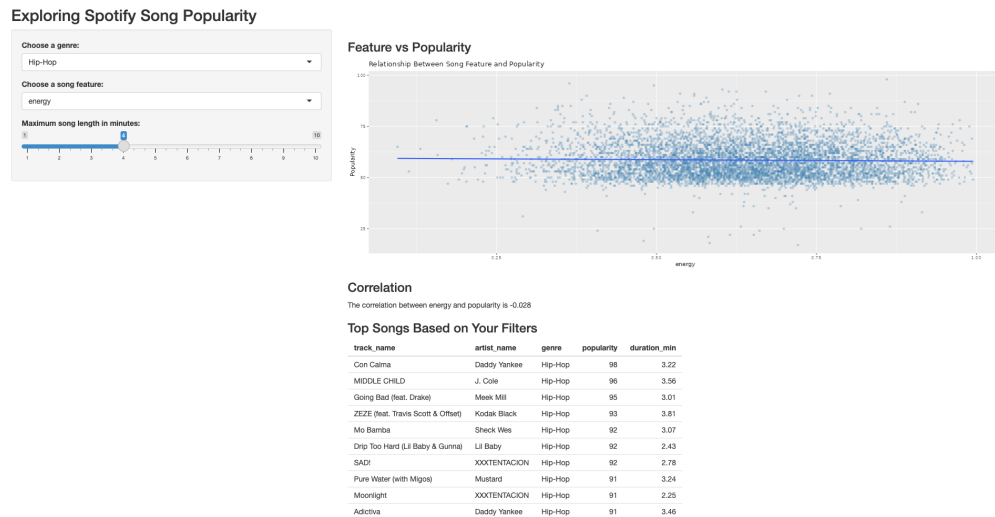


Figure 1: Data App

Technical Report

One of the biggest challenges I faced during this project was deciding how to turn a large Spotify dataset into an application that was both useful and easy to understand. The dataset contained thousands of songs and many different variables, so I had to decide which features would be most meaningful for users to see and test. I experimented with different ideas and eventually focused on song popularity and audio characteristics such as danceability, energy, loudness, tempo, and duration. I also had to think about which user inputs would provide meaningful insights without making the app super complicated. Through trial and error, I settled on a genre selector, a song feature selector, and a song length filter, which gave users flexibility while keeping the app organized and easy to use.

Another challenge was deciding which variables from the Spotify dataset would be most useful for analysis. The dataset contained many variables, and not all of them were relevant to

my research question. To address this, I selected variables related to popularity and audio characteristics such as danceability, energy, loudness, tempo, acousticness, and duration. I also performed data wrangling by selecting only the variables needed for the project, filtering out songs with a popularity score of zero, and creating a new variable that converted song duration from milliseconds to minutes. This process helped make the data easier to work with.

I used AI during the project as a debugging tool. I asked AI questions troubleshooting errors. Some of the suggestions were helpful, especially when I encountered coding errors or was unsure how to implement specific Shiny components. Instead of directly using every suggestion, I evaluated whether I could understand and explain the code myself. As a result, the final application reflects my own decisions about the dataset, variables, user inputs, and overall design while still benefiting from AI as a support tool.

Overall, this project helped me become more comfortable with data wrangling, visualization, and building a shiny app.